Agile Development Process and Devops Lab

Lab Assignment-03

Name: P MANUSH ROYAL

Reg.No:23MIS0418

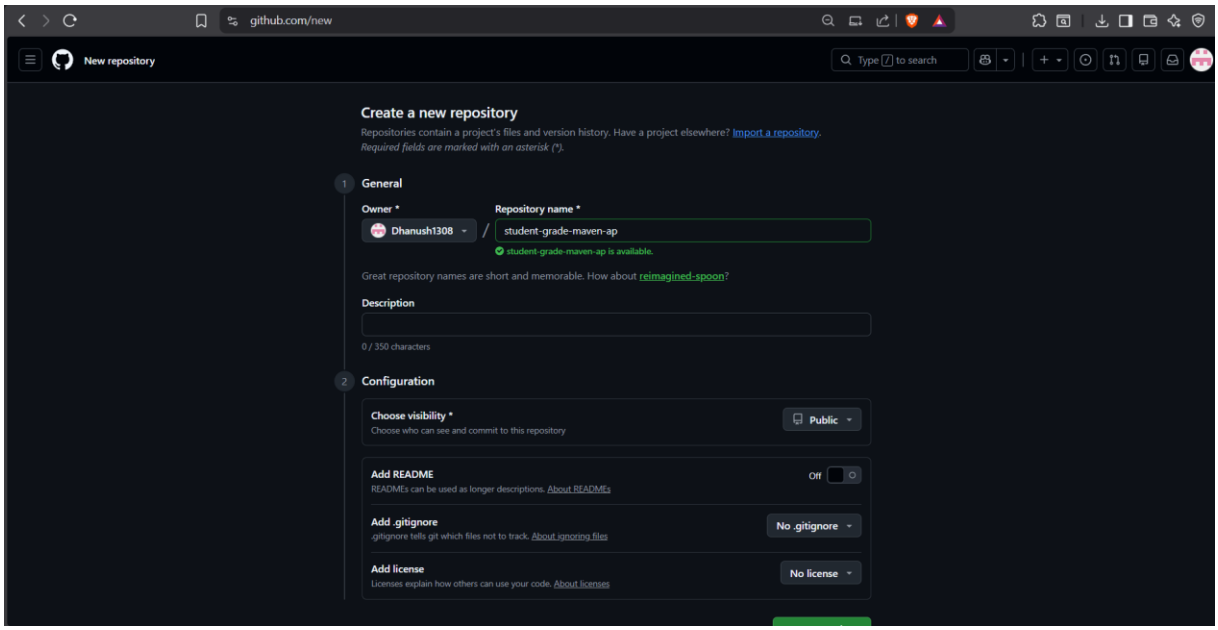
Slot:L3+L4

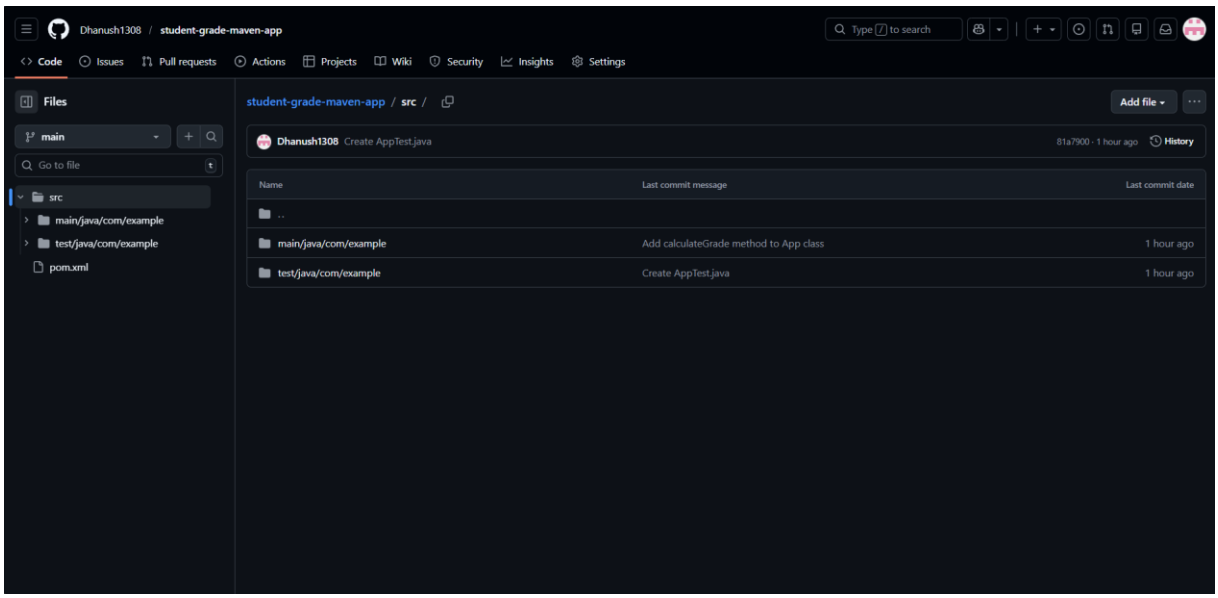
Faculty: SWETA BHATTACHARYA

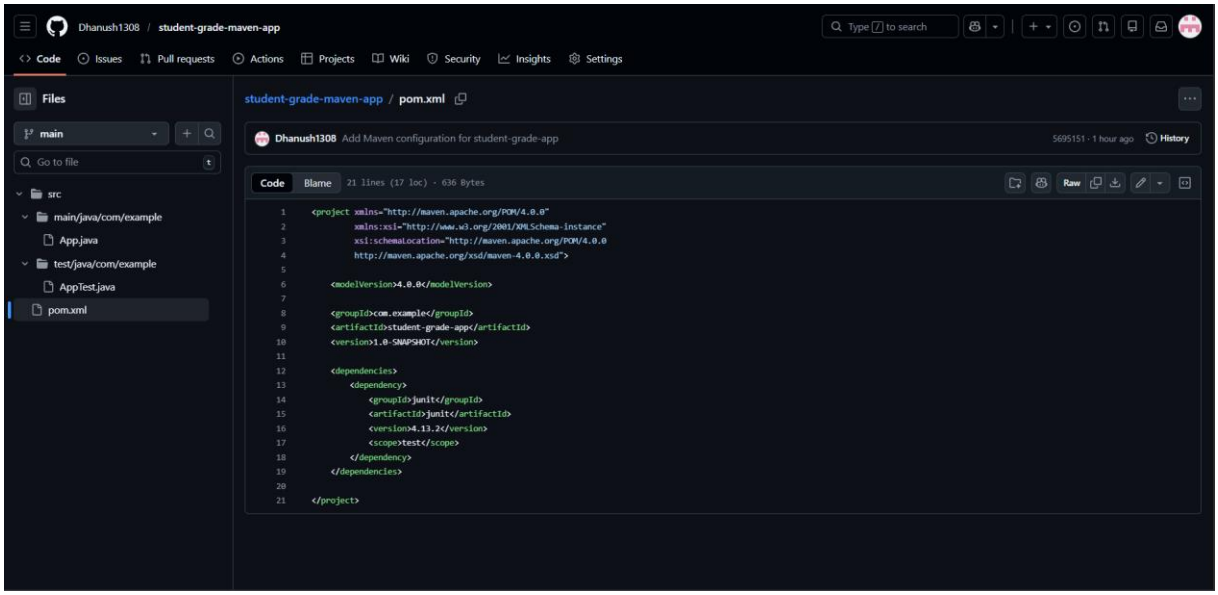
Course Code: ISWE406P

1. **Create a Maven Project in Git**

- Create a new Git repository
- Create a simple Maven project structure manually, including the following files and folders:
  - pom.xml
  - src/main/java/com/example/App.java
  - src/test/java/com/example/AppTest.java

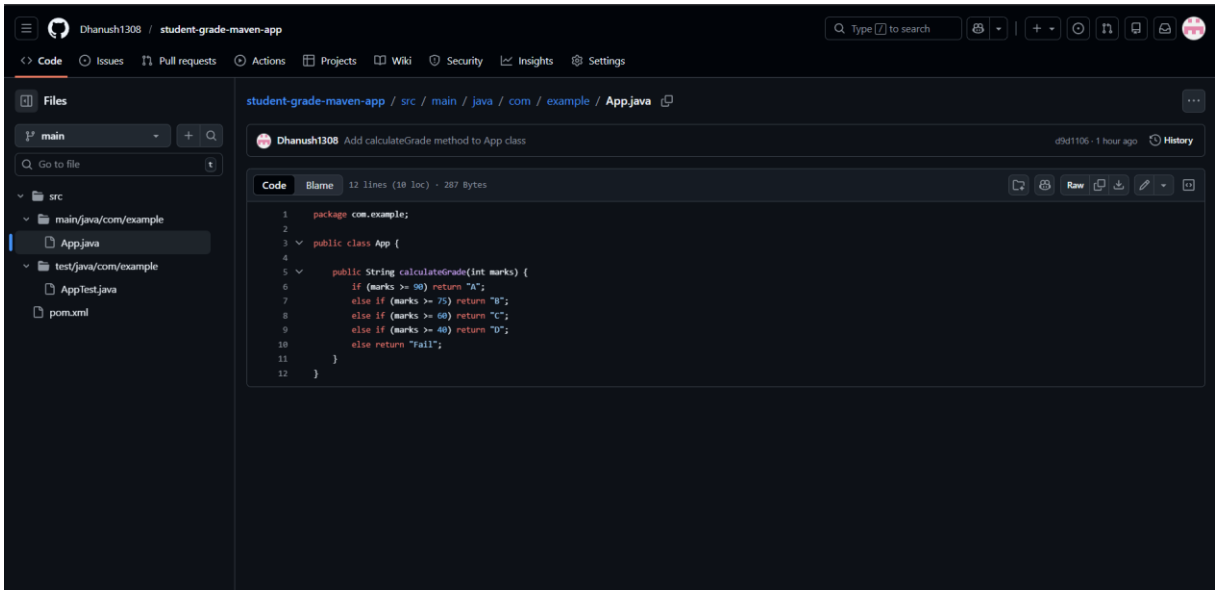


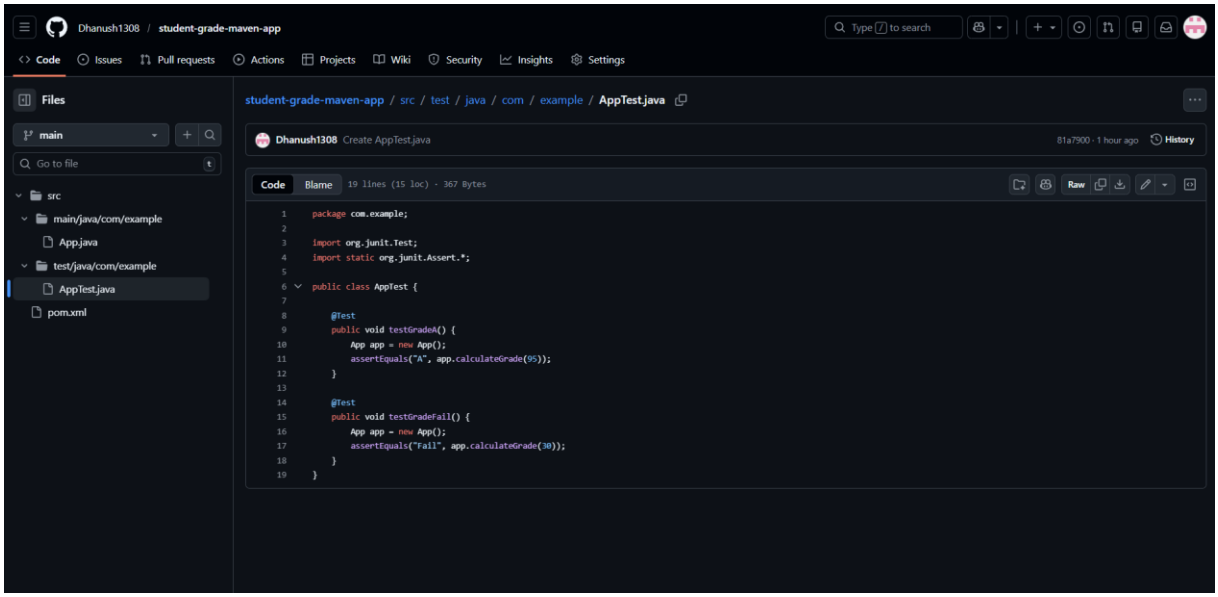




2. Implement the Java Program

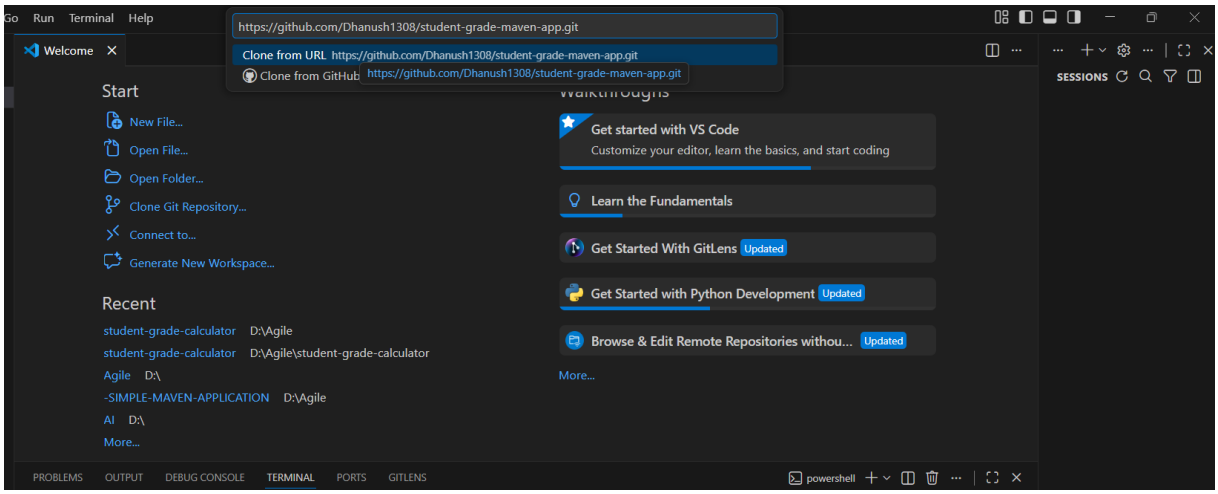
- Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios:
    - Student Grade Calculator
    - Write a corresponding unit test in AppTest.java.





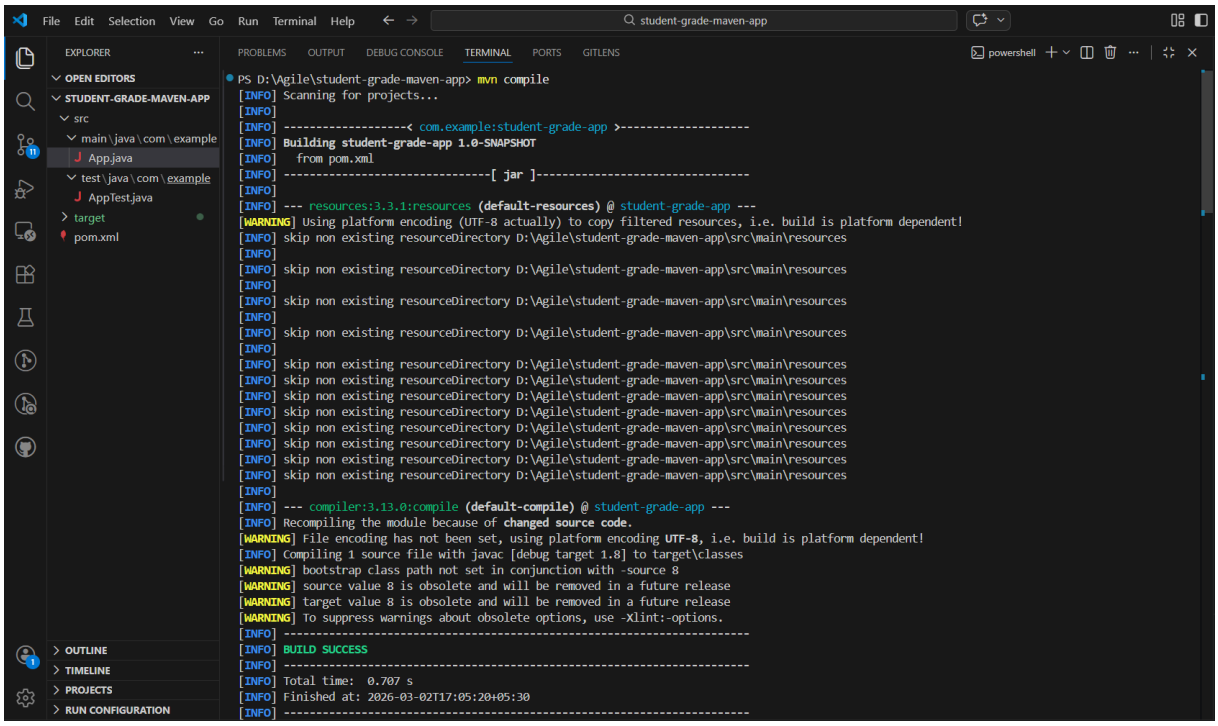
3. Push and Clone the Repository

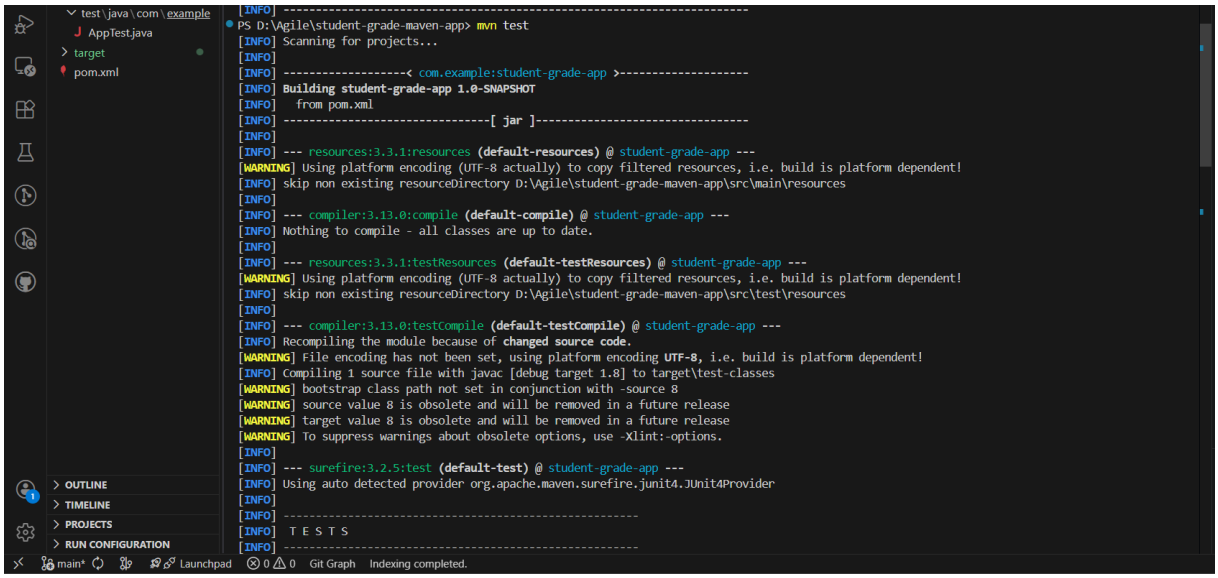
- Clone the repository using Visual Studio Code.
- Verify the project structure.



4. Build the Maven Project

- Use Maven commands in Visual Studio Code terminal to:
  - Compile the project
  - Run tests
  - Build the project (mvn clean install)





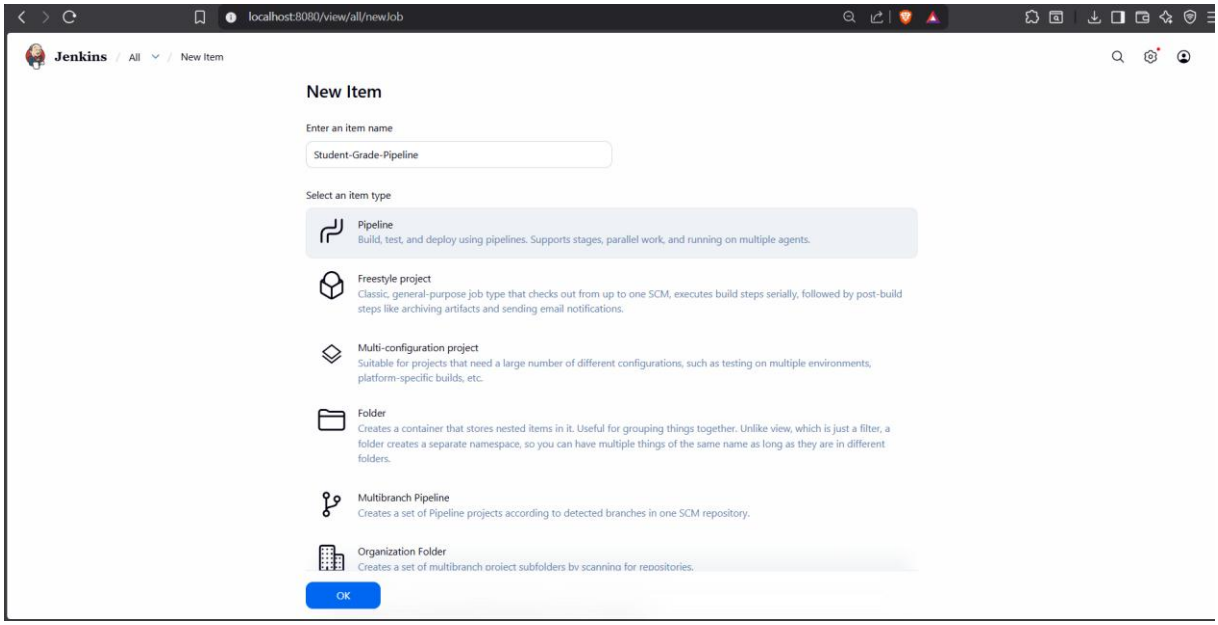
```
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.041 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.685 s
[INFO] Finished at: 2026-03-02T17:06:03+05:30
[INFO] ------------------------------------------------------------------------
PS D:\Agile\student-grade-maven-app> mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:student-grade-app >--------------------
[INFO] Building student-grade-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.4.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.4.1.pom (7.8 kB at
22 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.4.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.4.1.jar (34 kB at 6
0 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ student-grade-app ---
[INFO] Deleting D:\Agile\student-grade-maven-app\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ student-grade-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\Agile\student-grade-maven-app\src\main\resources
[INFO]
```

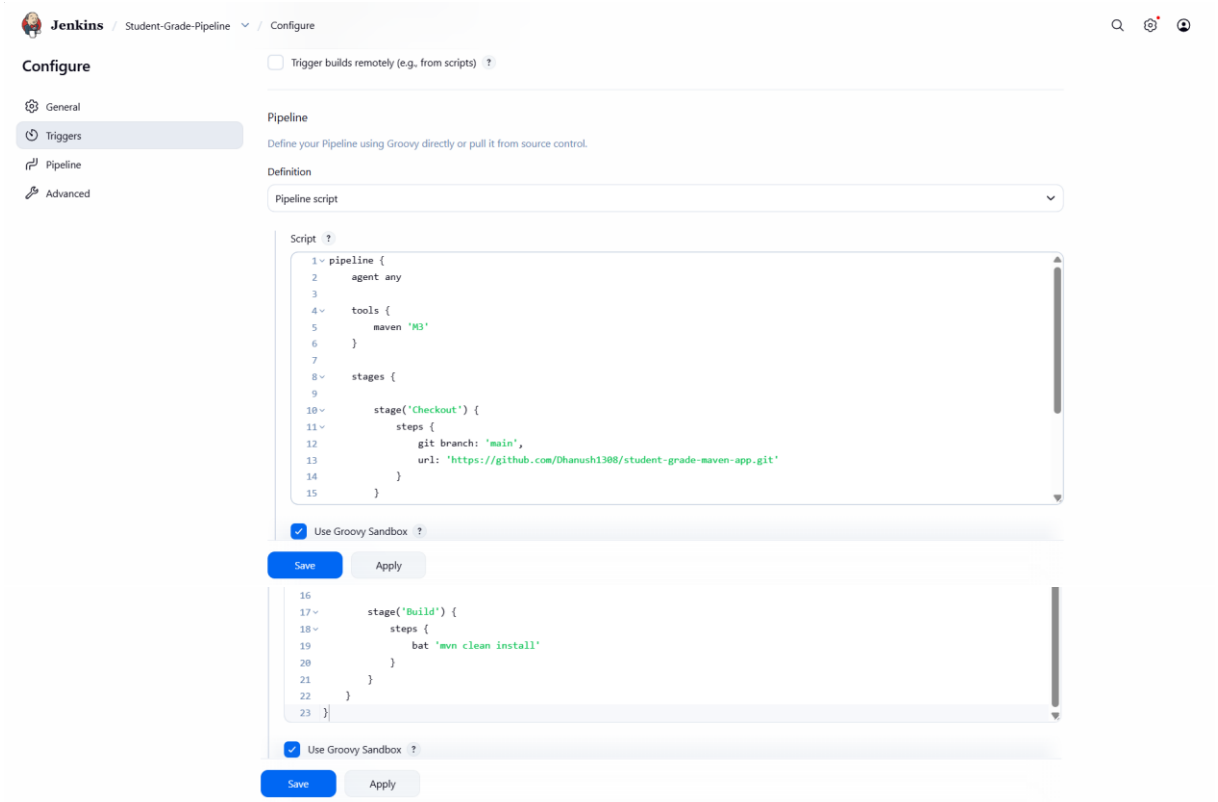
```
[INFO] --- compiler:3.13.0:compile (default-compile) @ student-grade-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ student-grade-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\Agile\student-grade-maven-app\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ student-grade-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ student-grade-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.039 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ student-grade-app ---
[INFO] --- jar:3.4.1:jar (default-jar) @ student-grade-app ---
```

```
PS D:\Agile\student-grade-maven-app> mvn clean install
[INFO] Building jar: D:\Agile\student-grade-maven-app\target\student-grade-app-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ student-grade-app ---
[INFO] Installing D:\Agile\student-grade-maven-app\pom.xml to C:\Users\death\.m2\repository\com\example\student-grade-app\1.0-SNAPSHOT\student-grade
-app-1.0-SNAPSHOT.pom
[INFO] Installing D:\Agile\student-grade-maven-app\target\student-grade-app-1.0-SNAPSHOT.jar to C:\Users\death\.m2\repository\com\example\student-gr
ade-app\1.0-SNAPSHOT\student-grade-app-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.616 s
[INFO] Finished at: 2026-03-02T17:06:23+05:30
[INFO] ------------------------------------------------------------------------
PS D:\Agile\student-grade-maven-app>
```

5. Create and Run Jenkins Pipeline

- Create a new Pipeline project in Jenkins.
- Create a pipeline script
- Execute the pipeline and verify successful build completion.





Jenkins / Student-Grade-Pipeline

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Stages
- Rename
- Pipeline Syntax

Builds

Filter /

Today

#1 5:08 PM

Student-Grade-Pipeline

Add description

Stage View

	Declarative: Tool Install	Checkout	Build
Average stage times: (full run time: ~20s)	90ms	1s	18s
#1 Mar 02 17:08 No Changes	90ms	1s	18s

Permalinks

- Last build (#1), 1 hr 34 min ago
- Last stable build (#1), 1 hr 34 min ago
- Last successful build (#1), 1 hr 34 min ago
- Last completed build (#1), 1 hr 34 min ago

REST API Jenkins 2.541.1

Jenkins / Student-Grade-Pipeline / #1

```
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\Student-Grade-Pipeline\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ student-grade-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ student-grade-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.045 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ student-grade-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.1.0/file-management-3.1.0.pom
Progress (1): 794 B
Progress (1): 2.2 kB
Progress (1): 4.4 kB
Progress (1): 4.5 kB
```

Jenkins / Student-Grade-Pipeline / #1

```
Progress (1): 163/196 kB
Progress (1): 180/196 kB
Progress (1): 196/196 kB
Progress (1): 196 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 494
kB/s)
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Student-Grade-Pipeline\pom.xml to
C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\example\student-grade-app\1.0-SNAPSHOT\student-grade-app-1.0-SNAPSHOT.pom
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Student-Grade-Pipeline\target\student-grade-app-1.0-SNAPSHOT.jar to
C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\example\student-grade-app\1.0-SNAPSHOT\student-grade-app-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  16.853 s
[INFO] Finished at: 2026-03-02T17:08:21+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.1